



Data Article

A multi-firearm, multi-orientation audio dataset of gunshots

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ARTICLE INFO

Article history:

Received 16 January 2023

Revised 15 February 2023

Accepted 20 March 2023

Available online 25 March 2023

Dataset link: [Gunshot/Gunfire Audio Dataset \(Original data\)](#)

Keywords:

Gunshot audio classification

Audio forensics

Machine learning

Acoustic situational awareness

Multiple sensor orchestration

Internet of Battlefield Things (IoBT)

ABSTRACT

Early detection of firearm discharge has become increasingly critical for situational awareness in both civilian and military domains. The ability to determine the location and model of a discharged firearm is vital, as this can inform effective response plans. To this end, several gunshot audio datasets have been released that aim to facilitate gunshot detection and classification of a discharged firearm based on acoustic signatures. However, these datasets often suffer from a lack of variety in the orientations of recording devices around the source of the gunshot. Additionally, these datasets often suffer from the absence of proper time synchronization, which prevents the usage of these datasets for determining the Direction of Arrival (DoA) of the sound. In this paper, we present a multi-firearm, multi-orientation time-synchronized audio dataset collected in a semi-controlled real-world setting – providing us a degree of supervision – using several edge devices positioned in and around an outdoor firing range.

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Specifications Table

Subject	Data Science / Applied Machine Learning
Specific subject area	Gunshot Audio Classification
Type of data	Audio recordings (Waveform Audio File Format (WAV), monophonic, 44.1kHz) Audio recording labels and metadata (Comma-Separated Values (CSV))
How the data were acquired	Audio recordings were acquired via a variety of edge devices using a variety of microphones as shown in Table 2.
Data format	Raw, Analyzed
Description of data collection	Audio was recorded during the firing sequence of four different types of firearms. Each round of firing was performed with three different styles: (1) Single shot fire; (2) Multiple shot fire where individual shots were separated in time; and (3) Rapid fire where each firearm was fired as quickly as possible.
Data source location	Institution: Eastern Florida State College's Firing Range at the Brevard County Sheriff's Office Work Farm City: Cocoa State: Florida Country: United States of America (USA) General Latitude and Longitude: 28.34214, -80.78145
Data accessibility	Repository name: Zenodo Data identification number: 10.5281/zenodo.7004819 Direct URL to data: https://zenodo.org/record/7004819

Value of the Data

- Provides gunshot audio collected from an outdoor firing range along with timestamps for gunshot occurrences within each file, the make and model of the firearm that fired the gunshot(s), the combinations of device types and microphones used to collect the audio, and their respective locations in the firing range.
- Data was collected under noisy real-world conditions. Extraneous sounds include sounds from a nearby highway, people speaking, wildlife sounds, and other outdoor environmental sounds typically found in real-world deployments. Additionally, since it was recorded in an outdoor firing range, the audio contains minimal echo.
- Can be used to train DoA estimation algorithms to perform in real-world conditions.
- Suitable for validating existing solutions for gunshot detection and localization.

1. Objective

There has been an increasing interest in the characterization of gunshots in both military and civilian settings, especially within the domain of cyber-physical Measurements And Signature IN-Telligence (MASINT) [1]. To the end of gunshot characterization, this manifests primarily as two tasks, namely, firearm classification and detection, and estimation of DoA for gunshots, which falls under a broader umbrella of gunshot localization methods. These tasks prove essential to law enforcement and military applications, where situational awareness can help formulate an effective response plan. Of late, there has been a relatively large corpus of literature on gunshot classification and detection [2–5], as well as the determination of DoA for gunshot localization using deep learning models [6,7]. Given the importance of the task at hand, as well as a re-

quirement to perform adequately well in previously unseen situations, high-quality and robust datasets are paramount to ensure reasonably effective performance in real-world situations.

AudioSet¹ constitutes a compilation of data from free online sources such as Freesound², SoundBible³, and YouTube. Morehead *et al.* [4] similarly include data collected from a firing range; however, it is difficult to ascertain the physical locations of the sensor device(s). Singh *et al.* [8] show that deep-learning models trained on the UrbanSound⁴ dataset for gunshot detection find it challenging to distinguish between plastic bag explosions and gunshots, which they remedy by training with a collection of bag-popping sounds. A false positive identification of gunshots can prove costly to law enforcement who need to dispatch officers per Ratcliffe *et al.* [9]. Another similar dataset was published by Cadre Forensics⁵, where they provide audio collected in a rural setting. However, this dataset cannot be used for DoA estimation since there is no guarantee that the audio signals were synced in time – even a time difference of a few milliseconds can throw off DoA estimations.

These aforementioned limitations in the datasets necessitate improvements to these datasets in the hope of improving the accuracy and robustness of gunshot alerting machine learning models in the presence of noise. With our present dataset, we address several of these limitations and contribute a high-quality dataset collected in a semi-controlled real-world setting. It is suggested that it be enriched with the previously mentioned datasets to improve the robustness of deep-learning models for gunshot detection and localization, especially for deployment in mission critical environments. Our dataset is especially useful for works that combine gunshot detection and localization – for instance, see [10,11].

2. Data Description

2.1. Dataset

This dataset consists of 2,148 audio files capturing gunshots fired by four selected firearms of interest, a CSV labels file, and a CSV summary file. The former CSV file contains the audio file name, the number of gunshots contained within the file, and the estimated timestamps of gunshot occurrences within the file (in seconds). The latter CSV contains the audio file names, the number of gunshots contained within each file, the make and model of the firearm that fired the gunshots, the time when the device started recording this audio, the manufacturer and model of the device used to record the audio along with a unique identifier for that particular device, the latitude and longitude of the device that recorded the audio, and finally, the timestamps of gunshot occurrences within the file in seconds. Gunshots are treated as having a single corresponding timestamp that was estimated using peak amplitudes. For instance, a recording containing four gunshots contains four corresponding timestamps representing the time audio for that gunshot was first received by the device. More details on this preprocessing step can be found in Section 3.

3. Experimental Design, Materials and Methods

3.1. Data collection setting

This data was collected at Eastern Florida State College's firing range (Cocoa, Florida, USA). The range is located with a highway to the west, a body of water surrounded by a wooded

¹ https://research.google.com/audioset/dataset/gunshot_gunfire.html

² <https://freesound.org>

³ <https://soundbible.com>

⁴ <https://www.kaggle.com/datasets/chrisfilo/urbansound8k>

⁵ <http://cadreforensics.com/audio/>

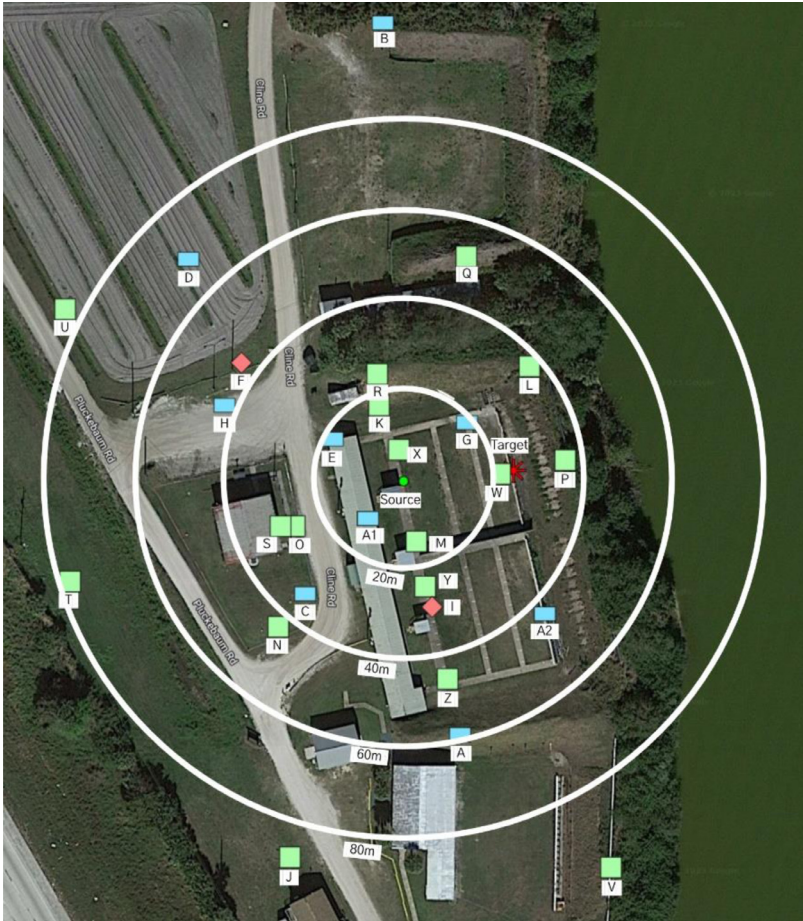


Fig. 1. Session 1 (all devices set to record): Each device and microphone combination is indicated by a colored rectangle. Blue rectangles indicate smartphones, while green squares indicate Raspberry Pis. Additionally, the WiFi hotspots used to create a mesh network are each indicated by a red diamond. Finally, the green dot indicates the source of the gunfire, while the dark red star-shaped symbol represents the target that was aimed towards during firing. Specifications regarding each device can be found in [Table 2](#).

area to the east, and law enforcement training environments to the north and south. The range itself is divided by a brick wall. Each half of the range has dimensions 85ft wide x 75ft deep. The north half of the range was used. Weapons were handled by a Florida Department of Law Enforcement (FDLE) state-certified firearms instructor assuming a standing position. The location of the instructor is shown in [Figs. 1 and 2](#). This setting provided an ideal combination of semi-controlled and real-world conditions for data collection. Its location at an outdoor firing range facilitated the collection of data amidst ambient noise, while its relative seclusion allowed some level of control over the type and amount of noise recorded by each device. For example, devices positioned closer to the west side of the range recorded a higher amount of highway related noise, while devices positioned closer to the east side of the range recorded a higher amount of woodland related noise.

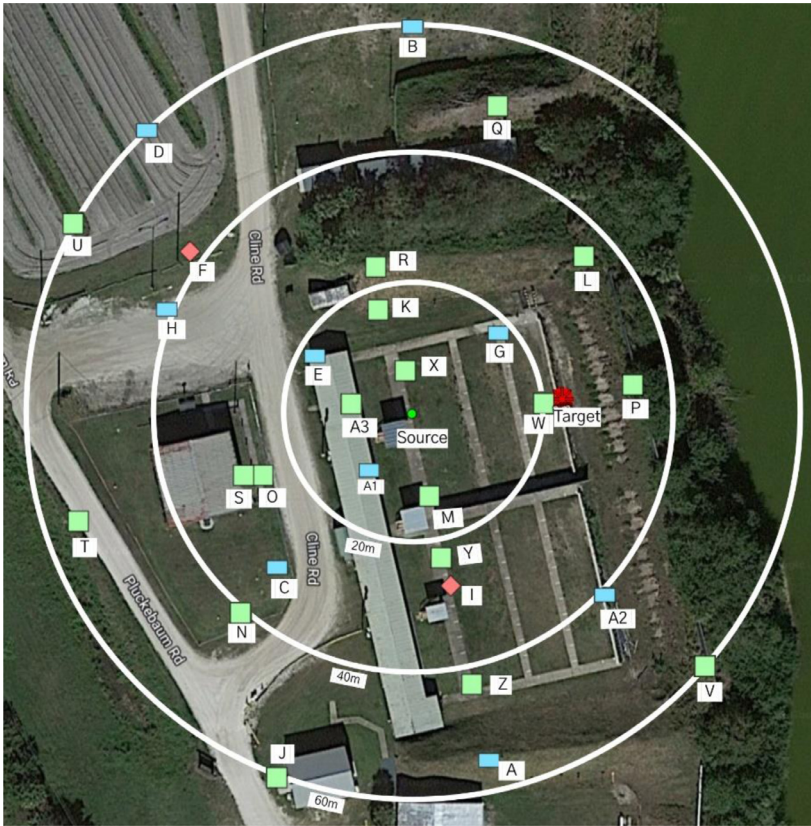


Fig. 2. Session 2 (most devices set to record): Colored shapes have the same meanings as in Fig. 1. Devices J, T, X, Z were set to perform inference (for firearm classification) during this session. Specifications regarding each device can be found in Table 2.

3.2. Materials

Four different types of firearms were fired. A detailed description of the firearms used is present in Table 1. Each firearm was shot in three different ways: (1) single shot fire (a single shot was fired), (2) multiple shot fire (multiple shots were fired with rest time in-between) (3) rapid fire (multiple shots were fired with minimal time in-between). For the rapid fire method, shots were fired as quickly as possible given that firearm type. For example, rapid fire from the Remington 870 would be much less “rapid” than that from other firearms due to the weapon’s pump-action nature. Each event was recorded by a series of edge devices with varying types of microphones. The position of these devices for each session is shown in Figs. 1 and 2. Table 2 describes the combinations of devices and microphones used.

3.3. Methods

3.3.1. Recording sessions

Two sessions of recording were conducted. During session 1, all devices were set to record. During session 2, four of the devices were set to perform inference to test models that were trained from data collected during session 1. Since these devices were performing inference and

Table 1
Firearms and Description, Firearm descriptors are listed in order of: manufacturer and model, type, relevant caliber or gauge depending on the firearm, and the number of files associated with each.

Image	Description
	Ruger ArmaLite Rifle (AR)-556 AR-15 style semiautomatic rifle .223 caliber 597 files
	Remington 870 Pump-action shotgun 12 gauge 379 files
	.38 Smith & Wesson Special Revolver .38 caliber 503 files

(continued on next page)

Table 1 (continued)

Image	Description
	Glock 17 Semi-automatic pistol 9mm caliber 669 files

Table 2
Device models used, individual named devices of each model and the microphones attached to each.

Device Manufacturer	Device Model	Device Name	Microphone Attached
Raspberry Pi	3B+ Zero W	A3	MOVO Dom2-USB Mini
		J	MiniDSP UMA-8 USB V2
		T	Changeek Gooseneck USB
		W	Galvanox USB-C Lavier
		X	ReSpeaker 2-Mics Pi Hat
		Y	ReSpeaker 2-Mics Pi Hat
		Z	MiniDSP UMA-8 USB V2
		K	Galvanox USB-C Lavier
		L	Changeek Gooseneck USB
		M	Galvanox USB-C Lavier
		N	Galvanox USB-C Lavier
		O	Bonke BM-858 Condenser
		P	MiniDSP UMA-8 USB V2
		Q	MiniDSP UMA-8 USB V2
		R	MiniDSP UMA-8 USB V2
		S	MiniDSP UMA-8 USB V2
		U	MiniDSP UMA-8 USB V2
		V	MiniDSP UMA-8 USB V2
Samsung	Galaxy S21	H	Internal
		B	Internal
		A1	Internal
	Galaxy S8	A	Internal
		D	Internal
		E	Internal
	Galaxy S9	C	Internal
		A2	Internal
		G	Internal

not recording for this session, the dataset contains fewer samples recorded by these devices than samples recorded by other devices.

3.3.2. Sound collection mobile application

A crucial feature of data collection is the feasibility of assigning a ground truth label to each data point. This endeavor increases in complexity when multiple geographically dispersed de-



Fig. 3. A screenshot of the sound collection application used for coordinating devices, assigning ground truth, and monitoring device status. The buttons showcased are responsible for sending MQTT messages to all registered devices. Note that some devices list their status as “mic error”, informing the user that a problem has occurred with the device’s connection to the microphone. In this way, the user can know exactly what state a device is in at any given moment.

vices are used for data collection. To reduce the amount of manual labelling performed post-collection, a dedicated sound collection mobile application was developed and used to orchestrate data collection and labeling. This application is shown in Fig. 3. This application allowed the researchers to assign ground truth on-site as data was collected and it used Message Queuing Telemetry Transport (MQTT) [12] as the message passing protocol to monitor device status, to instruct devices to record and stop recording, to instruct devices to either save a recording and its metadata or to discard a recording and all metadata, and to instruct devices to upload a recording and its metadata to the cloud. In order to facilitate this method, two WiFi hot spots were set up within the range. Ground truth for a recording consisted of ontology labels (“gun-shot”, model of firearm, caliber of firearm), and the number of gunshots fired during the recording.

4. Preprocessing

Preprocessing was performed on the collected audio to extract only the audio around the gunshot sounds. This was a multi-step process. First, signals were labeled automatically by comparing the “peak” amplitudes of the audio to the Root Mean Square (RMS) of the whole audio file and labeling any point above an empirically determined threshold of 5 times the RMS of the entire file as a gunshot. Then, these labels were manually verified by a human listener who corrected any non-gunshot sounds (for example, loud shouts, wind noise, etc.) that may have been mislabeled as gunshots and adjusted the automatic labels as necessary. An example of this gunshot event labeling schema and extraction for a clean audio file is shown in Fig. 4. An example of a waveform that required manual verification is shown in Fig. 5. Once the gunshot “peaks”

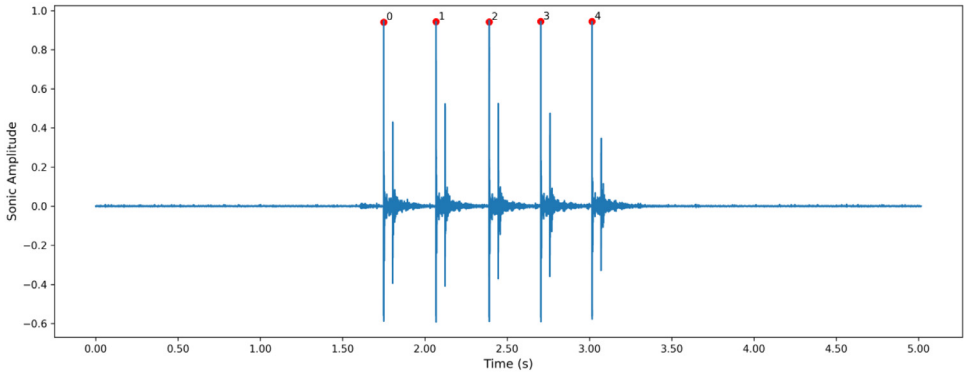


Fig. 4. Example of gunshot event extraction on file “41692623-c231-44d4-867a-1089f722a583_v0.wav”. In this situation 5 separate gunshots are observed with little ambiguity in where each gunshot begins.

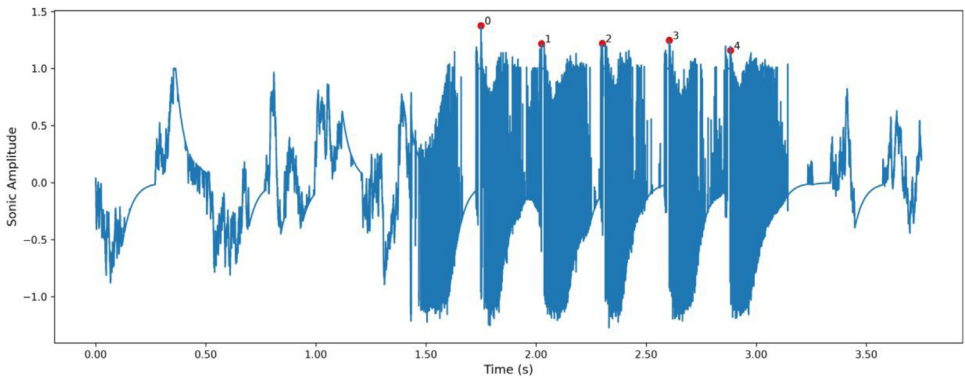


Fig. 5. Example of challenging gunshot event extraction on file “8ffbfb4-7f3b-456b-8d21-c64c9dfd943f_chan1_v0.wav”. In this situation, external noise interferes with our typical preprocessing scheme. Manual verification by a human listener is required to determine how many gunshots occur in the file and where exactly each gunshot begins.

were found and labeled, an additional step was applied to ensure that only one gunshot was labeled for every 0.25 seconds of audio (the observed average time between multiple shots). The audio was then separated into multiple clips by finding the best window that ensured that every 2 seconds of audio contained at least one gunshot (of length 0.25 seconds). If there was a gap larger than that amount, then the file was separated into multiple audio clips and the data in between was discarded. Each extracted audio clip was saved as its own file in the dataset. Finally, the final preprocessed audio files were manually verified again by another human listener. This extra step was taken to ensure that the final audio files were within an acceptable range of quality, preserving audio files containing real-world noise but excluding audio files where the gunshot was completely obscured by noise.

5. Working with Multi-Channel Audio for DoA

Eight 8-channel circular microphone arrays were used during the recording sessions. Raw audio recorded by the microphone arrays is synchronized to produce multi-channel recordings that can be used for DoA estimation. The specific microphone array used in our collection was

the MiniDSP UMA-8 USB mic array V2.0⁶. See MiniDSP's documentation⁷ for more detail about these microphone arrays. The recordings from these microphones are provided in the dataset both as their 8-channel WAV files and also as multiple individual mono WAV files. Information regarding identifying these files is present in the README for the dataset. There are many tools for manipulating multi-channel audio. In Python, a WAV file streamed using the PyAudio⁸ package can be split into eight channels by simply reshaping the audio frame. Alternatively, each channel can also be accessed as an individual waveform using a tool such as Audacity⁹. Once split into eight channels, the channel identifier corresponds to the microphone identifier on the microphone array.

Ethics Statement

This work did not involve human subjects, animal experiments, or data collected from social media platforms.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

[Gunshot/Gunfire Audio Dataset \(Original data\)](#) (Zenodo)

CRediT Author Statement

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Acknowledgments

We would like to thank Eastern Florida State College's range master Frank Ramos and Just Cause Tactical Training manager and instructor Scott Lindsley for their help in building this dataset. This material is based upon work supported by Air Force Research Laboratory (AFRL) grant No. FA8650-21-C-1147. Any opinions, findings, conclusions, or recommendations contained

⁶ <https://www.minidsp.com/products/usb-audio-interface/uma-8-microphone-array>

⁷ <https://www.minidsp.com/images/documents/UMA-8%20v2%20User%20manual.pdf>

⁸ <https://people.csail.mit.edu/hubert/pyaudio/>

⁹ <https://www.audacityteam.org>

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